

Lancaster University College
at Beijing Jiaotong University

Digital Signal Processing

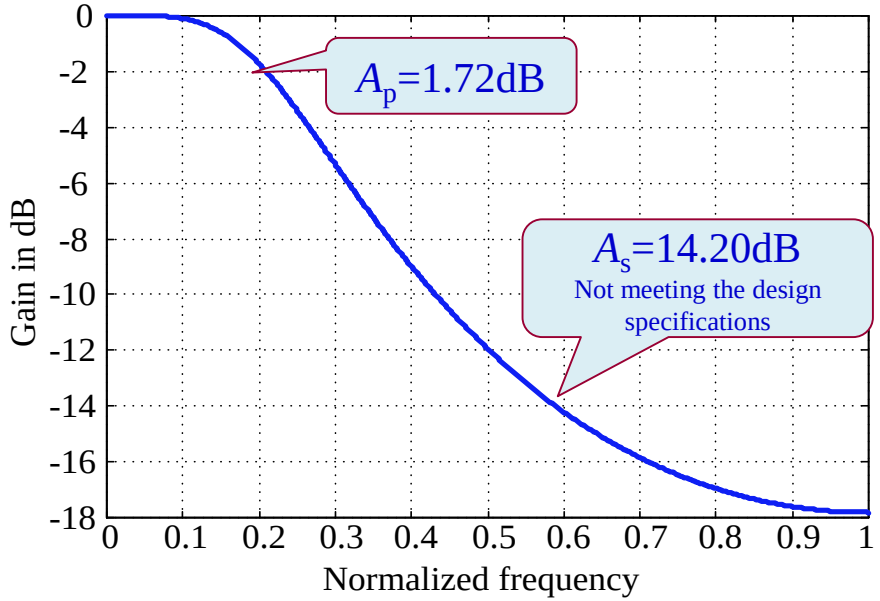
Zhou Hang

Bilinear Transformation

- Question
- Design principle
- Design steps

Question

Using BW type analog low-pass filter and impulse response invariant method to design and meet the specifications Ω_p=0.2π rad, Ω_s=0.6π rad, A_p≤2dB, A_s≥15dB digital low-pass filter.



Pulse response invariant method suffers from spectral aliasing

Solution

1.Increase the stopband attenuation of the DF to be designed

2.Adopting bilinear transformation method

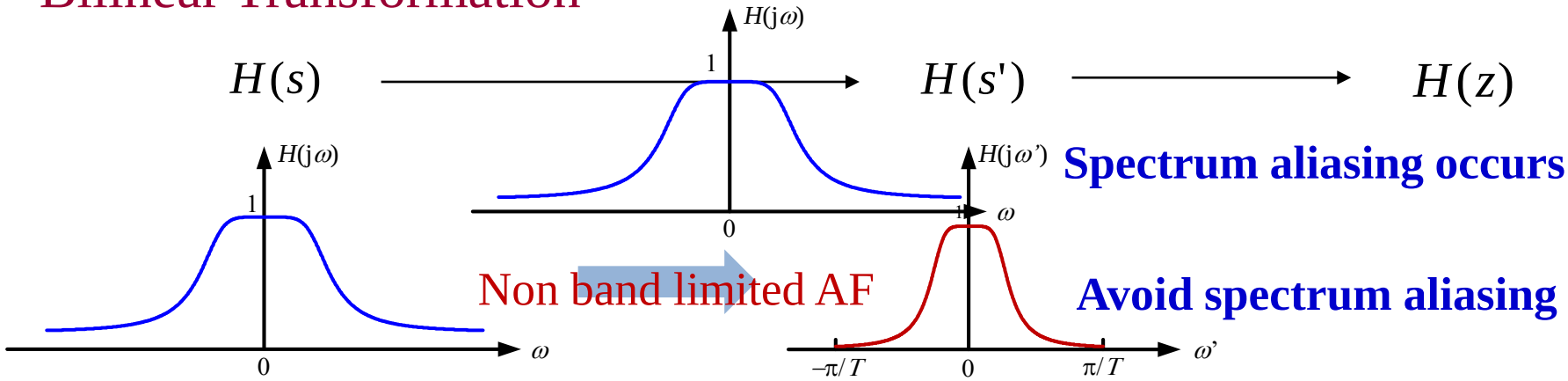
Design Principle

Impulse Invariance

$$H(s) \longrightarrow H(z)$$

Bilinear Transformation

$$H(s) \longrightarrow H(s') \longrightarrow H(z)$$



Design Principle

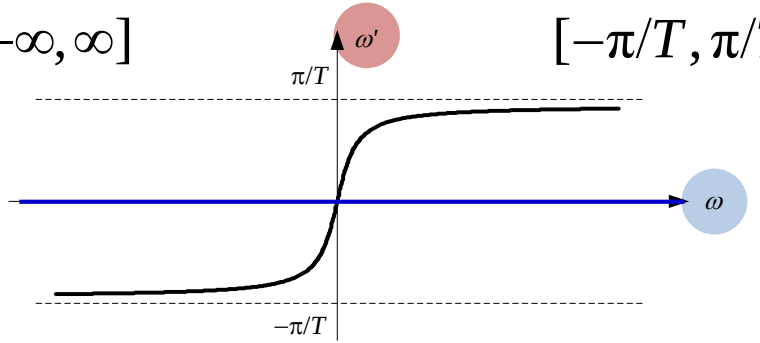
The basic idea of bilinear transformation:

Map non band limited analog filters to band limited analog filters with the highest angular frequency π/T.

$$H(s) \longrightarrow H(s') \longrightarrow H(z)$$

$$\omega \xrightarrow{\omega' = \frac{2}{T}\arctan(\frac{\omega T}{2})} \omega' \xrightarrow{\Omega = \omega' T} \Omega$$

$$[-\infty, \infty] \qquad [-\pi/T, \pi/T]$$



The relationship between analog frequency and digital frequency is

$$\Omega = 2\arctan(\frac{\omega T}{2})$$

$$\omega = \frac{2}{T}\tan(\frac{\Omega}{2})$$

Design Principle

The mapping relationship from the *s* domain to the *z* domain

$$\omega = \frac{2}{T}\tan\left(\frac{\Omega}{2}\right) \longrightarrow \mathrm{j}\omega = \mathrm{j}\frac{2}{T}\tan\left(\frac{\Omega}{2}\right) = \mathrm{j}\frac{2}{T}\frac{\sin\left(\frac{\Omega}{2}\right)}{\cos\left(\frac{\Omega}{2}\right)}$$

$$= \frac{2}{T}\frac{\mathrm{e}^{\mathrm{j}\frac{\Omega}{2}} - \mathrm{e}^{-\mathrm{j}\frac{\Omega}{2}}}{\mathrm{e}^{\mathrm{j}\frac{\Omega}{2}} + \mathrm{e}^{-\mathrm{j}\frac{\Omega}{2}}} = \frac{2}{T}\frac{1-\mathrm{e}^{-\mathrm{j}\Omega}}{1+\mathrm{e}^{-\mathrm{j}\Omega}}$$

$$\mathrm{j}\omega = \frac{2}{T}\frac{1-\mathrm{e}^{-\mathrm{j}\Omega}}{1+\mathrm{e}^{-\mathrm{j}\Omega}} \xrightarrow[z=\mathrm{e}^{\mathrm{j}\Omega}]{s=\mathrm{j}\omega} s = \frac{2}{T}\frac{1-z^{-1}}{1+z^{-1}} \longrightarrow z = \frac{2/T+s}{2/T-s}$$

Bilinear transformation

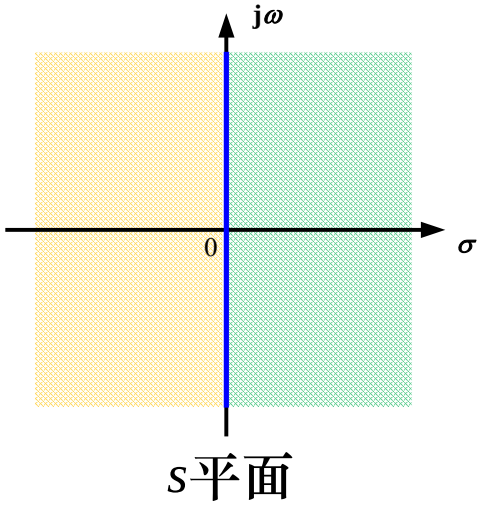
Design Principle

Stability analysis

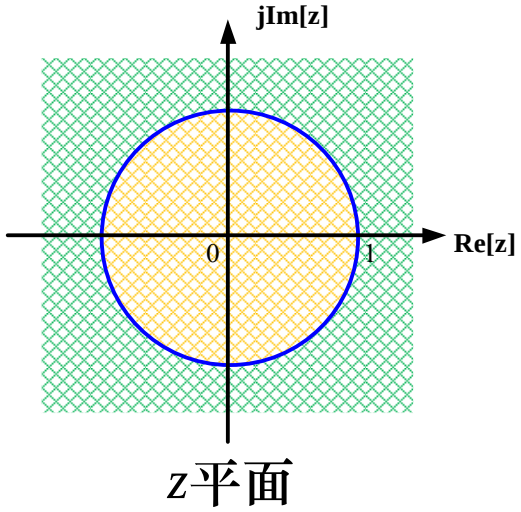
$s=0$ (s Plane imaginary axis) $\Leftrightarrow$ $|z|=1$ (On the unit circle of the z-plane)

$s<0$ (s Left-half plane) $\Leftrightarrow$ $|z|<1$ (z plane unit circle)

$s>0$ (s right-half plane) $\Leftrightarrow$ $|z|>1$ (z plane unit circle outside)



s平面



z平面

Design Principle

Advantages and disadvantages of bilinear transformation method

Advantages:

No spectrum aliasing.

Disadvantages:

The relationship between Ω and ω is not linear. If the frequency response of the analog filter is not a piecewise constant, amplitude distortion will occur.

Design Principle

1. Convert the frequency index $\{\Omega_i\}$ of DF to the frequency index $\{\omega_i\}$ of AF

$$\omega_i = \frac{2}{T}\tan(\frac{\Omega_i}{2})$$

2. Design the $H(s)$ of the analog filter based on its specifications

3. Using bilinear transformation, convert $H(s)$ to $H(z)$.

$$H(z) = H(s)\Big|_{s=\frac{2}{T}\frac{1-z^{-1}}{1+z^{-1}}}$$

[E.g] Design a BW type analog low-pass filter and bilinear transformation method to meet the requirements Ω_p=0.2π rad, Ω_s=0.6π rad, A_p≤2dB, A_s≥15dB digital low-pass filter

Solution: (1) Convert digital low-pass indicators into analog low-pass indicators, and take T=2 。

$$\omega_p = \frac{2}{T}\tan(\frac{\Omega_p}{2}) = 0.3249, \quad \omega_s = \frac{2}{T}\tan(\frac{\Omega_s}{2}) = 1.3764,$$

$$\omega_i = \frac{2}{T}\tan(\frac{\Omega_i}{2})$$

A_p≤2dB, A_s≥15dB

(2) Design BW type analog low-pass filter

$$N \geq \frac{\lg\ (\frac{10^{0.1A_s}-1}{10^{0.1A_p}-1})}{2\lg(\omega_s/\omega_p)} = 2 \qquad \omega_c = \frac{\omega_s}{(10^{0.1A_s}-1)^{1/(2N)}} = 0.5851$$

$$H_L(s) = \frac{1}{(\frac{s}{\omega_c})^2 + \sqrt{2}\frac{s}{\omega_c} + 1} = \frac{0.3423}{s^2 + 0.8275s + 0.3423}$$

[E.g] Design a BW type analog low-pass filter and bilinear transformation method to meet the requirements $\Omega_p=0.2\pi$ rad, $\Omega_s=0.6\pi$ rad, $A_p\leq 2$dB, $A_s\geq 15$dB digital low-pass filter

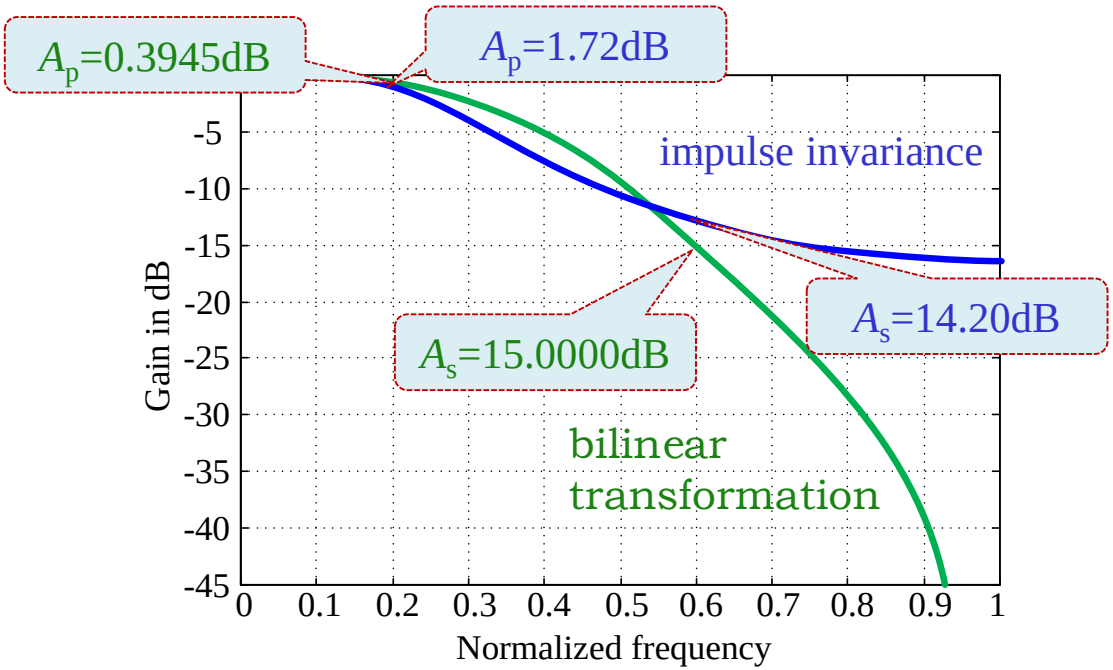
Solution: (3) Convert an analog low-pass filter into a digital low-pass filter using bilinear transformation method

$$H(z)=H(s)\Big|_{s=\frac{2}{T}\frac{1-z^{-1}}{1+z^{-1}}}$$

$$=\frac{0.1578+0.3155z^{-1}+0.1578z^{-2}}{1-0.6062z^{-1}+0.2373z^{-2}}$$

$$H(e^{j\Omega})=H(z)\Big|_{z=e^{j\Omega}}$$

Pulse response invariant method vs bilinear transformation method



Technical specifications of low-pass digital filter:

Ω_p=0.2π rad, Ω_s=0.6π rad,

A_p≤2dB, A_s≥15dB

The impulse response invariant method suffers from spectral aliasing, and the designed DF does not meet the given criteria;

The bilinear transformation method does not have spectral aliasing, and the designed DF meets the given criteria.

[E.g] **Design BW type digital band stop filter using bilinear transformation method that meets the following criteria**

$\Omega_{p1}=0.90\pi$ rad, $\Omega_{p2}=0.95\pi$ rad, $A_p\leq 1$dB,

$\Omega_{s1}=0.93\pi$ rad, $\Omega_{s2}=0.94\pi$ rad, $A_s\geq 10$dB

Solution: (1) Convert digital bandstop filter indicators to analog bandstop filter indicators

Set $T=2$, known $\omega=\frac{2}{T}\tan(\frac{\Omega}{2})$

The simulated bandstop filter index obtained is:

$\omega_{p1}=6.3$ rad/s, $\omega_{p2}=12.7$ rad/s, $A_p\leq 1$dB

$\omega_{s1}=9.1$ rad/s, $\omega_{s2}=10.6$ rad/s, $A_s\geq 10$dB

[E.g] **Design BW type digital band stop filter using bilinear transformation method that meets the following criteria**

Ω_{p1}=0.90π rad, Ω_{p2}=0.95π rad, A_p≤1dB,

Ω_{s1}=0.93π rad, Ω_{s2}=0.94π rad, A_s≥10dB

Solution: (2) Convert the simulated bandstop filter index to the simulated low-pass filter index

$$B = \omega_{s2} - \omega_{s1} = 1.5 \quad \omega_0 = \sqrt{\omega_{s1}\omega_{s2}} = 9.7889$$

$$\overline{\omega}_p = \max\{\left|\frac{B\omega_{p1}}{-\omega_{p1}^2 + \omega_0^2}\right|, \left|\frac{B\omega_{p2}}{-\omega_{p2}^2 + \omega_0^2}\right|\} = 0.2945$$

$$\overline{\omega}_s = 1 \qquad A_p \le 1\text{dB}, A_s \ge 10\text{dB}$$

[E.g] **Design BW type digital band stop filter using bilinear transformation method that meets the following criteria**

Ω_{p1}=0.90π rad, Ω_{p2}=0.95π rad, A_p≤1dB,

Ω_{s1}=0.93π rad, Ω_{s2}=0.94π rad, A_s≥10dB

Solution: (3) Design prototype BW type analog low-pass filter

$$N \geq \frac{\lg\left(\frac{10^{0.1A_s}-1}{10^{0.1A_p}-1}\right)}{2\lg\left(\bar{\omega}_s/\bar{\omega}_p\right)} = 2 \qquad \omega_c = \frac{\bar{\omega}_s}{(10^{0.1A_s}-1)^{1/(2N)}} = 0.5774$$

The system function of the prototype simulated low-pass filter is

$$H_L(\bar{s}) = \frac{1}{\left(\frac{\bar{s}}{\omega_c}\right)^2 + \sqrt{2}\frac{\bar{s}}{\omega_c} + 1} = \frac{0.3333}{\bar{s}^2 + 0.8165\bar{s} + 0.3333}$$

[E.g] **Design BW type digital band stop filter using bilinear transformation method that meets the following criteria**

$\Omega_{p1}=0.90\pi$ rad, $\Omega_{p2}=0.95\pi$ rad, $A_p\leq 1$dB,

$\Omega_{s1}=0.93\pi$ rad, $\Omega_{s2}=0.94\pi$ rad, $A_s\geq 10$dB

Solution: (4) Convert the prototype simulated low-pass to simulated band stop filter through complex frequency transformation

$$H_{\mathrm{BS}}(s)=H_{\mathrm{L}}(\bar{s})\Big|_{\bar{s}=\frac{Bs}{s^2+\omega_0^2}}=\frac{s^4+198s^2+9801}{s^4+4.899s^3+210s^2+485s+9801}$$

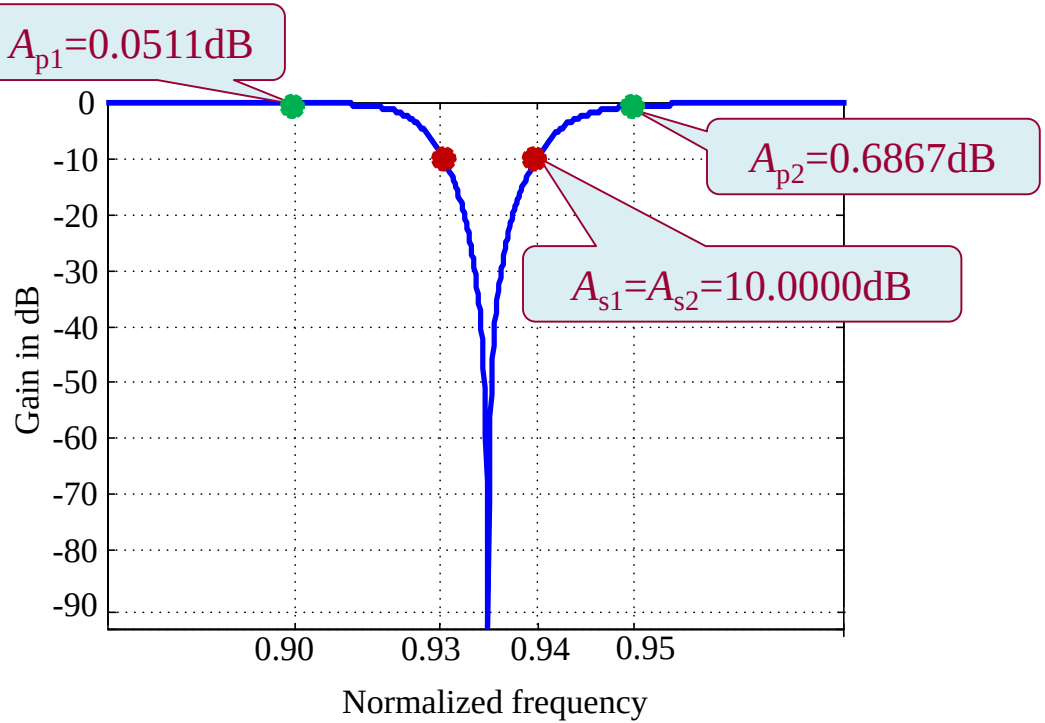
The impulse response invariant method cannot be used to design digital band stop filters, so the bilinear transformation method is adopted.

(5) Convert analog bandstop filter from bilinear transformation to digital bandstop filter

$$H(z)=H(s)\Big|_{s=\frac{2}{T}\frac{1-z^{-1}}{1+z^{-1}}}=\frac{0.9522+3.7327z^{-1}+5.5624z^{-2}+3.7327z^{-3}+0.9522z^{-4}}{1+3.8241z^{-1}+5.5601z^{-2}+3.6412z^{-3}+0.9067z^{-4}}$$

[E.g] **Design BW type digital band stop filter using bilinear transformation method that meets the following criteria**

$\Omega_{p1}=0.90\pi$ rad, $\Omega_{p2}=0.95\pi$ rad, $A_p \leq 1$dB,
$\Omega_{s1}=0.93\pi$ rad, $\Omega_{s2}=0.94\pi$ rad, $A_s \geq 10$dB



Convert analog filter to digital filter

bilinear transformation

[numd,dend] = bilinear(num,den,Fs)

num,den： Coefficient vectors of AF numerator and denominator polynomials

Fs=1/T： sampling frequency

numd,dend： Coefficient vectors of DF numerator and denominator polynomials

[E.g] Design a BW type analog low-pass filter and bilinear transformation method to meet the requirements Ω_p=0.2π rad, Ω_s=0.6π rad, A_p≤2dB, A_s≥15dB digital low-pass filter

```
%Design DF BW low-pass filter
%DF BW LP specification
Wp=0.2*pi; Ws=0.6*pi; Ap=2; As=15;
T=2; Fs=1/T;
%Analog Butterworth specification
wp=tan(Wp/2)*2/T; ws=tan(Ws/2)*2/T;
%determine the order of AF filter
[N,wc]=buttord(wp,ws,Ap,As,'s');
%determine the AF-BW filter
 [numa,dena]=butter(N,wc,'s');
%determine the DF filter
 [numd,dend]=bilinear(numa,dena,Fs)
```

```
%plot the frequency response
 w=linspace(0,pi,1024);
 h=freqz(numd,dend,w);
%normalize amplitude
 norm=max(abs(h));
 plot(w/pi,20*log10(abs(h/norm)));
 xlabel('Normalized frequency');
 ylabel('Gain in dB');
 %computer Ap As of the designed filter
 w=[Wp Ws];
 h=freqz(numd,dend,w);
 fprintf('Ap= %.4f\n',-20*log10( abs(h(1))));
 fprintf('As= %.4f\n',-20*log10( abs(h(2))));
```

Thanks

The materials referenced in this course are accumulated over years of teaching by the instructor, sourced from various media and exchanges with colleagues and peers. It is difficult to indicate the source of each material one by one. This is to clarify and express gratitude.